

Solutions to large scale DNA and RNA processing

Yorick Molema, Daniël Scheepstra, Martijn Kruijer, Wesley Haverkort,
Jeroen Klooster, Julian Pasveer & Rutger Wuijster

Abstract—Big data sets of DNA and RNA can be easily acquired. However, processing these data-sets, which are in the petabyte scale, comes with some complications. This report will discuss how already existing computational solutions can help with processing large-scale data sets. The best solution found in this report is cloud computing: it improves the speed, flexibility and costs of processing data. However, it can compromise privacy. Map Reduce and Heterogeneous computing were the other solutions taken into consideration. Life sciences are in need of a solution, because if a solution is found, further research into the human body can be carried out. Furthermore, the size of these data sets will keep growing. This means when no solution can be found, the problem will become harder to solve.

I. INTRODUCTION

NOWADAYS you can sequence the DNA and RNA of an entire human being for less than US\$ 5.000 [1]. Rapid advancements in the life sciences enable small individual laboratories to generate large amounts of data at a reasonable cost. These extensive sets of data acquired can exceed the petabyte scale. These small laboratories don't have the computational infrastructure to process these huge data sets. At the moment they use solutions from other fields in science, but there is still room for a lot of innovation.[2] This means that new computational solutions are required to keep advancing in the life sciences. These new computational solutions would solve the large scale data management and analysis. Cloud computing and heterogeneous computing might be two of the best solutions [2] and will be thoroughly discussed in this paper. Cloud and heterogeneous computing are already being used in other sciences and are provided by the big tech companies such as Amazon, meaning that there is no need to set up a complete new platform, which makes seeking to a solution quite a lot easier.

The size of the data sets won't stop growing and in 5 - 10 years it will out grow the petabyte scale and enter the exabyte scale.[2] Due to this fact a solution is needed. Processing data of this scale might not be practically feasible, unless there would be a solution that is being worked on right now. Furthermore, if a solution to processing DNA and RNA data sets arises, new information of the human body can be obtained.

In this research paper the computational solutions to large scale data problems in life sciences will be discussed, after which the previously described methodologies used to solve the data problem will be discussed, how they can fit into the life sciences and how it will be used in the life sciences. In the end a view on the solutions and the problem is given.

II. METHODOLOGIES TO PROCESS DNA AND RNA DATA SETS

The problem being addressed is to compute the large data sets more efficiently. This is limited by money, power, space and people[2]. In order to compute the large scale data sets more efficiently we look at 3 techniques used by other sciences in science, namely: Cloud Computing, Map Reduce and Heterogeneous Computing.

A. Cloud computing

A possible way to tackle big data problems is by using cloud computing. High-Performance Computing (HPC) was done mainly by cluster-based computing. This is however very expensive to create and operate[2]. One way to circumvent these disadvantages is to use virtualization technology. Virtualization software allows creating a system that operates like a normal physical computer but has flexible specification, such as number of processors or memory size. The use of this virtual software, that is on-demand, is known as cloud computing.[3][4]

Internet-based companies, like Google, Amazon and Microsoft, are able to invest in 'mega-scale' computational clusters and advanced, large-scale data-storage systems. For example, this allows Amazon to launch an on-demand cloud computation with their Amazon EC2 On-Demand[5]. According to Amazon, this allows their users to free them from the costs and complexities of planning, purchasing, and maintaining hardware and transforms what are commonly large fixed costs into much smaller variable costs. This cost reduction is one of the biggest advantages of cloud computing. For around the same nominal cost you can speed up computations by running them parallel. Instead of running 1 day on 1 Central Processing Unit (CPU), you can run the computations on multiple CPU's to bring the running time down to 2 hours and pay the same. This on-demand system provides great flexibility. Another big benefit of cloud computing is the ease of transferring and sharing of data. Due to the fact that data sets and analysis results can be held in the cloud they can easily be shared. These shared data sets and results offer greater flexibility and reduces the overall cost.

The benefits come at a cost, namely one of the disadvantages of cloud computing is that you have less control over the hardware, transfer rates to and from the cloud and more. Cloud computing has advantages when data is already in the cloud but it does not tackle the problem to get the data from the "source" to the cloud.

Another big problem with cloud computing is the privacy concerns. When the cloud servers are publicly accessible it has to be in compliance with a lot of regulations. For example,

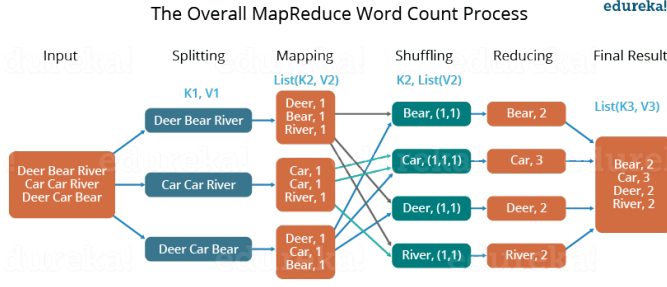


Fig. 1. MapReduce on a data-set to sort words [8]

when we want to store genomic information (DNA or RNA) there are plenty of laws and policies and there is ongoing debate as to whether further measures are needed [6].

B. Map reduce

Another solution for these data issues would be using MapReduce. MapReduce is a programming framework that allows for a lot of data to be processed at the same time by dividing the data in smaller chunks [7]. These chunks will then be evaluated individually in parallel. This basically means that the problem is divided into multiple smaller problems, which will later be connected to each other to get the desired output. First, the data is split into multiple blocks, after that the framework will pose the 'map' step. This step will produce key-value pairs after a certain block of data is processed. With this intermediate output, the key-value pairs, we can do the next step, which as the name suggests, is 'reduce'. Reduce will sort the key-value pairs out and reduce everything to one output. An example can be seen in figure 1, which is an illustration from Edureka[8], which shows the counting of words using MapReduce. This is great, because the entire process can run on thousands of different machines with the many sub-problems [7]. For example, when taking 10,000 cheap servers, each with 128MB data-blocks, 1TB of data can be processed simultaneously. This is easily scalable [7], because another 10,000 servers can be plugged in with larger data-blocks to increase the data processed. This shows that MapReduce can do a great deal in processing large sets of data.

This process is, in practice, used by different companies that use large amounts of data sets, which included Google until 2014 [9]. Google traded it in for an even more advanced version of MapReduce, which is less disk orientated and reduces latency [10].

A downside to MapReduce is that not every large-scale data-set is applicable with a basic MapReduce. Algorithms that are complex and that need specific, complex data-sets cannot use the MapReduce approach, but would need more attention to figure out how the data is accessed. For example, reconstructing a certain RNA for a special protein, would not benefit from the simplistic framework MapReduce provides. In this case, a different solution would be more beneficial.

C. Heterogeneous Computing

Instead of making use of cloud-based computing, one can use the so called heterogeneous multiple core computers. These computers have integrated specialized accelerators that increase peak arithmetic throughput by 10-fold to 100-fold and can turn individual computers, the work-horses of both desktop and cluster computers, into mini-supercomputers [11]. Almost all computer and laptops come with a Graphics Processing Unit (GPU), which is most commonly used for video game rendering, as the name suggests. However, it can also be used for genomic applications. GPU's are available for relatively low prices and offer so called 'cluster-scale' performance, which means that more than one trillion floating point operations (or calculations with decimal numbers) happen per second [2]. These are called FLOPS. GPU's might only contribute 5% of the processors, they do however contribute to 60% of all the FLOPS [12] [13]. GPU's are used for doing several operations on genomes. Algorithms are being developed for these GPU's as well to make all the computations possible.

Heterogeneous computing sounds promising. The increase of speed is sometimes fivefold or even tenfold and using a laptop instead of a dedicated workstation [2]. Due to these extraordinary improvements in speed, much costs are saved. Think about all the costs which go to data storage. The data can be stored locally and equipment used to maintain it is unnecessary. Besides this, GPU's already consist of great computational power, since their purpose is mostly for video game rendering, which asks a lot from a computer. The overall costs that are being saved need to be taken into consideration, since they do match up with the computational power that is required for an application which uses tightly coupled parallelism. Some application are undoubtedly useless when using the heterogeneous systems. As said in "Computational solutions to large-scale data management and analysis" [2], this is due to the fact that when, for example, constructing a Bayesian network with fewer than 25 nodes, the overhead is too great in relation to the amount of work done by the GPU, to achieve an advantage over a GPP. A GPP is a General Purpose Processor, which are often the processors that power desktop computers or workstations. Networks with more than 25 nodes can take great advantage of heterogeneous systems.

There is a downside to heterogeneous computing. A programmer needs to know how to develop or modify their applications which are implemented on GPU's. This is because it can get very complex. In some cases, algorithms need to be completely rewritten in order to run on a GPU. To be able to make use of the heterogeneous system, algorithms have to be modified or, as said before, completely rewritten which is very time consuming.

III. DISCUSSION

As is mentioned, there are different ways in which the big data processing problems can be tackled. The first of which is cloud computing. This methodology improves data processing in different respects. It improves the speed, flexibility and cost. The biggest problem that comes with the use of cloud computing are the privacy concerns, because the data could

accidentally or intentionally be disclosed to unauthorized individuals. Another problem, although it only applies when data is not yet stored in the cloud, is the fact that uploading data onto the cloud takes quite a lot of time.

The next methodology we will discuss is Map Reduce. This methodology improves big data processing with respect to the speed and cost. The biggest disadvantage is that some data sets cannot be processed unless there is a complex algorithm in place for the mapping and reducing of the data. Which results in the methodology being less efficient and most likely not the best way to solve the problem at hand.

Last, we're going to take a look at Heterogeneous Computing. Their main improvement here is raw speed. As said before their speed increases roughly of fivefold to tenfold, reducing the cost in the process. These Heterogeneous computers increase the arithmetic throughput by 10-fold or even 100-fold. However, the biggest disadvantage is that they need to modify the applications to implement these improvements.

IV. CONCLUSION

For processing large-scale data-sets the most appropriate option is using cloud computing. This can be attributed to the fact that cloud computing offers significantly more advantages compared to disadvantages than other methods. One of the biggest advantages for cloud computing is of course the flexibility. When different scales of computational power are required, this can easily be adjusted. Despite the disadvantage of having to upload these data-sets to the internet, which are often enormously large and may take days to upload, this is still a viable option. However since the laboratories producing the data that needs to be processed might be in more remote locations, which do not have the right internet stability to support uploading large files, there is still some other possibility for these laboratories. If a laboratory does not have great internet access, they might pick something like Heterogeneous Computing. Since this type of processing does not rely on the internet speed from the laboratory to a cloud server this might be a more variable option. Even with these things in mind, it might be possible that in the near future some algorithm, making the Map Reduce solution possible and easy, will be created. When this happens this might be the best solution to all options.

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